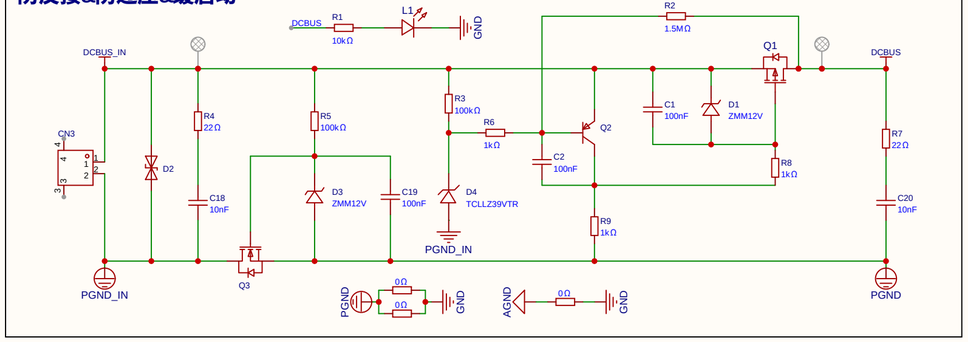
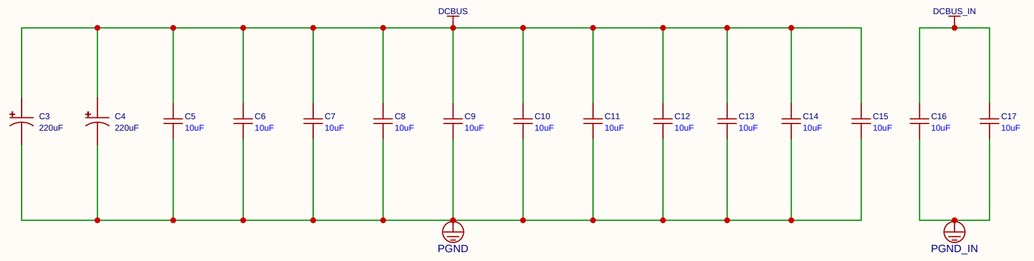
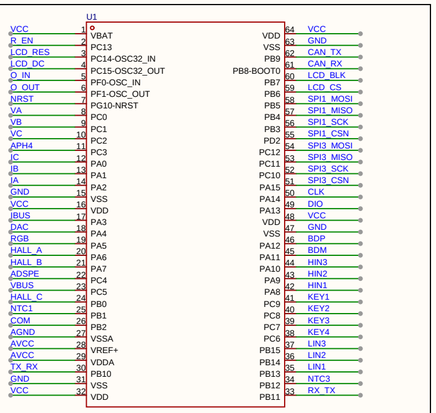
防反接&防过压&缓启动



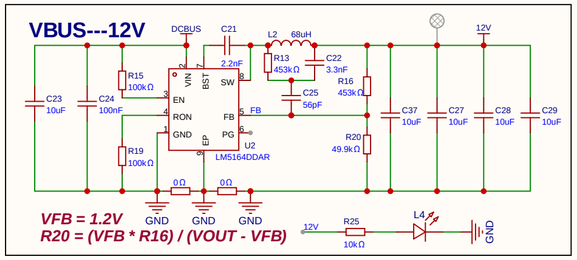
电源滤波



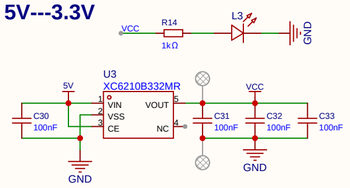
MCU



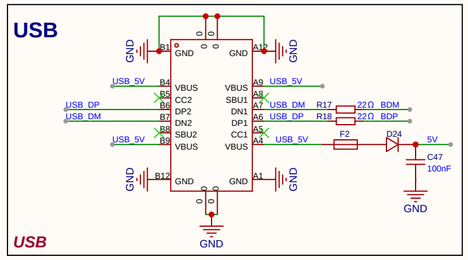
VBUS---12V



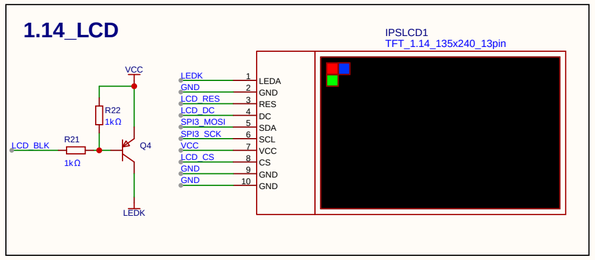
5V---3.3V



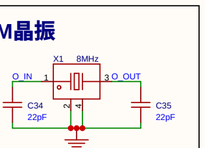
USB



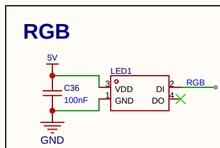
1.14_LCD



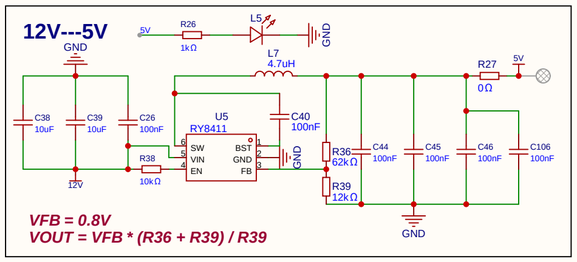
8M晶振



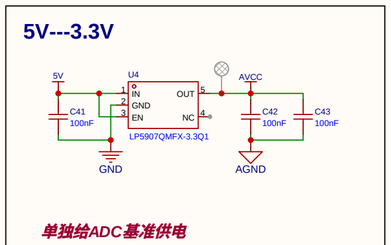
RGB



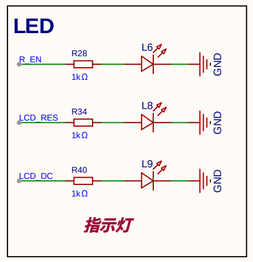
12V---5V



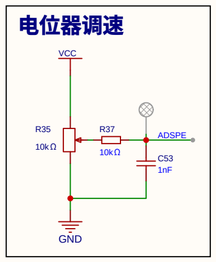
5V---3.3V



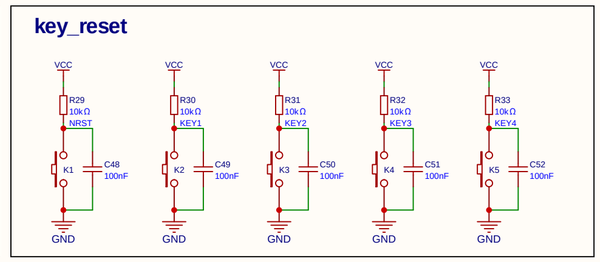
LED



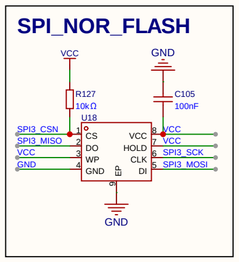
电位器调速



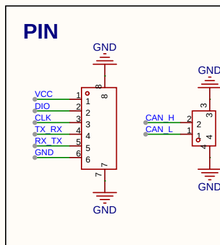
key_reset



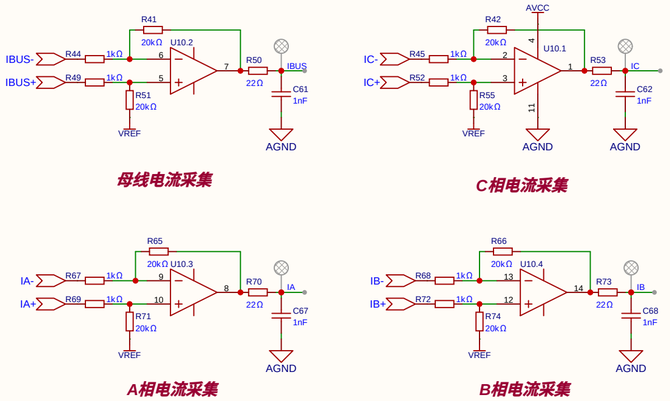
SPI_NOR_FLASH



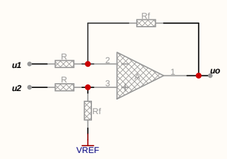
PIN



电流采样



电流采样计算



运放增益:k = Rf / R

偏置电压:VREF = 1.65V = 2048

采样电阻:Rm = 1mΩ

运放输出:uo = k * (u2 - u1)

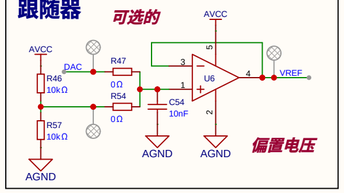
u2 - u1 = I * Rm

uo = (adc_val - 2048) * 3.3V/4096

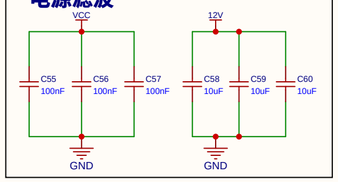
采集电流:I = uo / (Rm * k)

跟随器

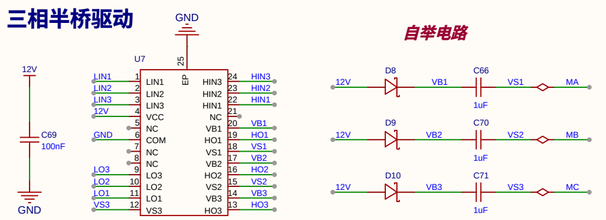
可选的



电源滤波

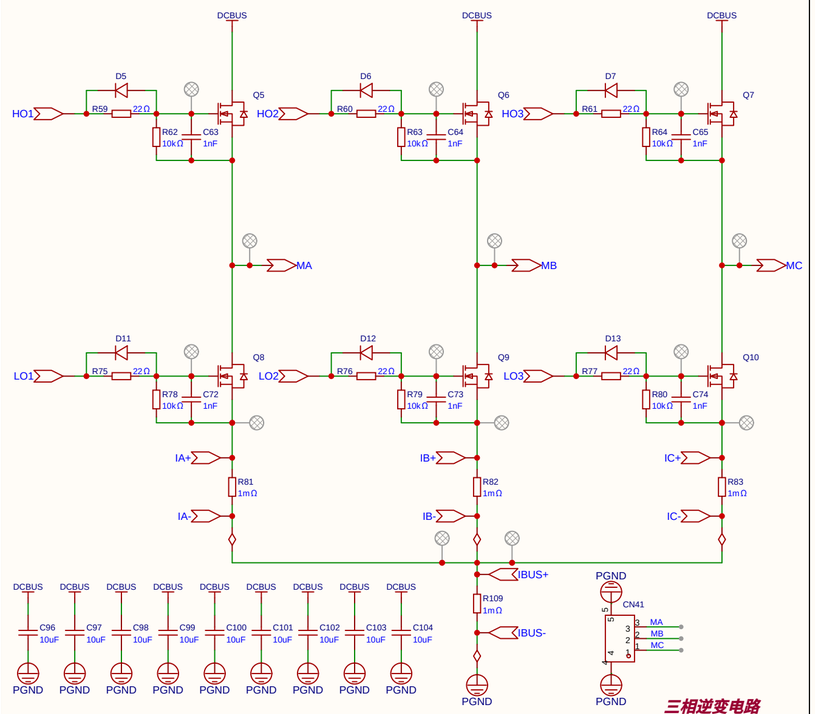


三相半桥驱动



自举电路

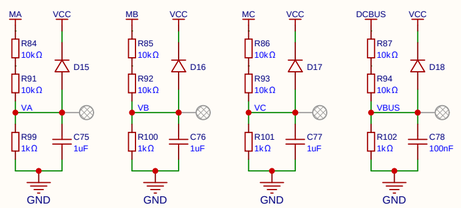
驱动电路



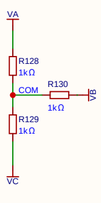
电压采样

167Hz 截止频率

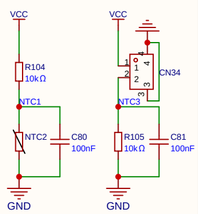
f=1/(2*pi*C76*((R85+R92)||R100))



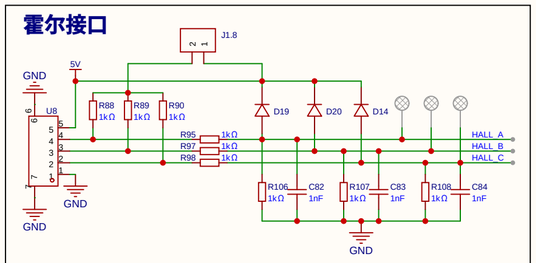
中性点采样



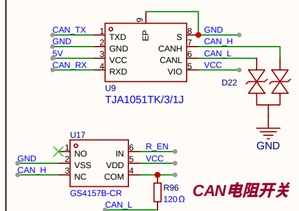
温度采样



霍尔接口



CAN通讯



采样点最高电压为: 3.3V + 二极管管压降

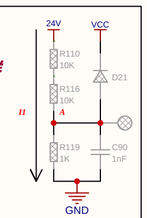
采样点电压应接近3.3V的一半,分辨率高，易采样

I1电流: 24V/(10k+10k+1k) = 1.14mA

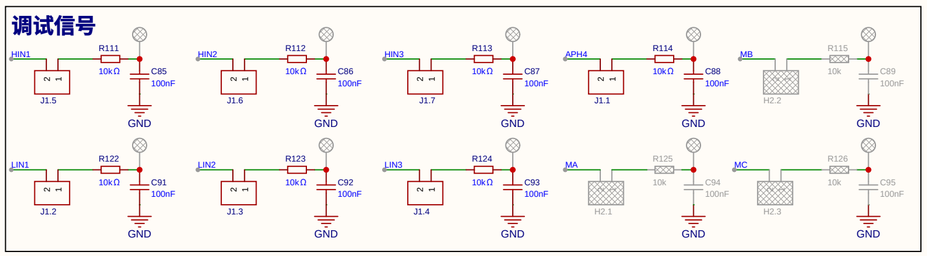
A点电压: I1 * 1k = 1.14V

缩小电压系数: k = (10k+10k+1k) / 1k

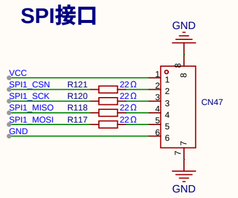
VA = adc_val * (3.3V/4096) * k



调试信号



SPI接口



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Page	P1	Create Date	2025-04-01
Drawed		Part Number	
Reviewed		FOCDriver	
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